

Zoravur Singh

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EDUCATION

Honours Bachelor of Computer Science

Major in Computer Science with Co-op, Minor in Pure Mathematics

University of Waterloo

2018 — 2023

SKILLS

Languages | Python, C, C++, Go, TypeScript

Frameworks & Libraries | React, Redux, NumPy, Pandas, JAX, Playwright, FastAPI

Infrastructure & Tools | PostgreSQL, sqlite, Docker, AWS, Snowflake

Systems & Concepts | Wasm, CMake, web scraping, machine learning

EXPERIENCE

Solutions That Matter

Toronto, ON

Software Engineer

Dec 2023 — Jun 2025

- Designed and built an order processing system end-to-end, automating \$1M+/year in orders and eliminating manual data entry errors.
- Developed a Python ETL service for automatic order reconciliation, reducing a 12+ hour manual task to minutes.
- Shipped an internal order tracking system in 3 days using Google Sheets, replacing back-and-forth of status inquiries.

DealMaker

Toronto, ON

Data Engineering Intern

May 2023 — Sep 2023

- Built analytics dashboards in Metabase using SQL to track sales performance, ad spend ROI, and customer metrics.
- Improved ETL pipeline performance 5x via batch API requests and extended it to support new data sources.
- Prototyped a machine learning model to forecast capital raise outcomes using historical transaction data.

Royal Bank of Canada

Toronto, ON

AI Engineering Intern

Sep 2021 — Apr 2022

- Contributed to Aiden, an RL-based trading algorithm handling \$100M+ daily, by architecting new trading signals.
- Developed a C++ market exchange simulation environment, reducing training cycles from 3 days to 14 hours.
- Built MLP classifiers to predict momentum vs. reversion market conditions, improving ETF trading performance.

Arcturus Networks

Toronto, ON

Software Engineering Intern

May 2020 — Aug 2020

- Built a video surveillance application using React, Redux, redux-saga, and ZeroMQ for real-time streaming.
- Implemented live alerts, notifications, and pipeline parameter configuration features.
- Reduced Docker build times from 7 minutes to 20 seconds by optimizing Dockerfiles and CMake cache usage.

ROSS Intelligence, Inc.

Toronto, ON

Machine Intelligence Engineering Intern

May 2019 — Aug 2019

- Extended backend microservices and refactored auth logic to allow users to use the app without creating an account, driving customer acquisition.
- Built a subscription settings page in React with functionality to modify, view, and cancel subscriptions.
- Migrated 1000 lines of the frontend codebase from JavaScript to TypeScript.

PROJECTS

Spock | A spoken programming language  zoravur.com/spock-a-spoken

- Designed and implemented a tree-walk interpreter for a programming language using English words without punctuation, and integrated OpenAI Whisper (speech-to-text) to enable voice-dictated programming.

Taan Generator | Indian classical music generator  (demo)

- Trained a transformer model to generate taans (melodic runs) in the Indian classical raga Yaman.